

Chapter 3

Supersymmetry, shape invariance and exceptional orthogonal polynomials

In the previous chapter, we explored fundamental mathematical concepts essential to our current discussion. EOPs play a crucial role in the rational extension of ES potentials, helping to expand the list of such potentials. In this chapter, we study the differential equations associated with EOPs from a different perspective. By adopting the approach of Ashok Das et al. [196, 197], we analyze these EOPs from an operator viewpoint in the SUSY framework. For simplicity, first we analyze X_1 -Jacobi and X_1 -Laguerre EOPs. Once we have established the foundational understanding, we extend our study to the more general case, namely X_m -Jacobi and X_m -Laguerre EOPs, for any integer value of m . To begin this comprehensive study, we start by reformulating the corresponding differential equations as eigenvalue equations. This step allows us to take advantage of the powerful SUSY framework. By doing this, we may investigate the underlying symmetries and structures that control these differential equations [229]. This approach not only helps in obtaining the solutions of these equations but also reveals the underlying SI and SUSY symmetries that make these EOPs particularly interesting and significant in mathematical physics.

3.1 X_1 -Exceptional Orthogonal polynomials

In this section, we discuss the X_1 -Jacobi and X_1 -Laguerre polynomials within the framework of SUSY QM and SI techniques, highlighting the reasoning behind the solubility of their corresponding differential equations.

3.1.1 X_1 -Jacobi Orthogonal polynomials

The differential equation satisfied by the X_1 -Jacobi orthogonal polynomials $\hat{P}_n^{(\alpha,\beta)}(x)$ are given in (2.25), where $n = 1, 2, 3, \dots$ and the real parameter $\alpha > -1$ and $\beta > -1$.

An eigenvalue equation can be used to represent Eq. (2.25),

$$\begin{aligned} & \left[(x^2 - 1) \frac{d^2}{dx^2} + \left((\beta + \alpha + 2)x - (\beta - \alpha) + \frac{2(\beta - \alpha)(x^2 - 1)}{(\beta + \alpha) - (\beta - \alpha)x} \right) \frac{d}{dx} \right. \\ & \left. + \left((\beta - \alpha)x + \frac{(\beta - \alpha)^2(x^2 - 1)}{(\beta + \alpha) - (\beta - \alpha)x} - (n - 1)(n + \alpha + \beta) \right) \right] \hat{P}_n^{(\alpha,\beta)}(x) = 0 \end{aligned} \quad (3.1)$$

$n = 1, 2, 3, \dots$

This can also be written as

$$[H - (n - 1)(n + \alpha + \beta)] \hat{P}_n^{(\alpha,\beta)}(x) = 0, \quad (3.2)$$

where

$$\begin{aligned} H = (x^2 - 1) \frac{d^2}{dx^2} &+ \left((\beta + \alpha + 2)x - (\beta - \alpha) + \frac{2(\beta - \alpha)(x^2 - 1)}{(\beta + \alpha) - (\beta - \alpha)x} \right) \frac{d}{dx} \\ &+ \left((\beta - \alpha)x + \frac{(\beta - \alpha)^2(x^2 - 1)}{(\beta + \alpha) - (\beta - \alpha)x} \right). \end{aligned} \quad (3.3)$$

Regarding the X_1 -Jacobi polynomials, the lowering $A_{\alpha,\beta}$ and raising $B_{\alpha,\beta}$ operators are defined as

$$A_{\alpha,\beta} = \frac{(x - c)^2}{x - b} \frac{d}{dx} \left(\frac{1}{x - c} \right), \quad (3.4a)$$

$$B_{\alpha,\beta} = (x^2 - 1) \frac{(x - b)}{(x - c)} \left(\frac{d}{dx} + a \right) - a(x^2 - 2bx + 1), \quad (3.4b)$$

with the following relations relating a , b , and c to α , β

$$a = \frac{\beta - \alpha}{2}, \quad (3.5a)$$

$$b = \frac{\beta + \alpha}{\beta - \alpha}, \quad (3.5b)$$

$$c = b + \frac{1}{a}. \quad (3.5c)$$

and these operators hold the following relations

$$A_{\alpha,\beta} \hat{P}_n^{(\alpha,\beta)}(x) = \left(\frac{n+\alpha+\beta}{2} \right) \hat{P}_{n-1}^{(\alpha+1,\beta+1)}, \quad (3.6a)$$

$$B_{\alpha,\beta} \hat{P}_n^{(\alpha+1,\beta+1)} = 2n \hat{P}_{n+1}^{(\alpha,\beta)}. \quad (3.6b)$$

These two operators can then be used to factorize the Hamiltonian in Eq. (3.3) as

$$H = B_{\alpha,\beta} A_{\alpha,\beta} = A_{\alpha-1,\beta-1} B_{\alpha-1,\beta-1} - (\alpha + \beta). \quad (3.7)$$

By using Eqs. (3.6a) and (3.6b) in Eq. (3.2) we have

$$\begin{aligned} B_{\alpha,\beta} A_{\alpha,\beta} \hat{P}_n^{(\alpha,\beta)} &= E_n \hat{P}_n^{(\alpha,\beta)} \\ A_{\alpha,\beta} B_{\alpha,\beta} (A_{\alpha,\beta} \hat{P}_n^{(\alpha,\beta)}) &= E_n (A_{\alpha,\beta} \hat{P}_n^{(\alpha,\beta)}), \end{aligned} \quad (3.8)$$

along with

$$E_n = (n-1)(n+\alpha+\beta). \quad (3.9)$$

As a result, H^1 and H^2 , as defined by

$$H^1 = B_{\alpha,\beta} A_{\alpha,\beta} \quad (3.10a)$$

$$H^2 = A_{\alpha,\beta} B_{\alpha,\beta} \quad (3.10b)$$

are SUSY partner Hamiltonians. In a more generalized way, we can demonstrate that the pair of Hamiltonians $B_{\alpha+k-1,\beta+k-1} A_{\alpha+k-1,\beta+k-1}$ and $A_{\alpha+k-1,\beta+k-1} B_{\alpha+k-1,\beta+k-1}$ are also the SUSY partner Hamiltonians (for $k = 1, 2, 3, \dots$). Eqs. (3.6a) and (3.6b), in particular, allow us to write,

$$B_{\alpha+k,\beta+k} A_{\alpha+k,\beta+k} \hat{P}_n^{(\alpha+k,\beta+k)} = E_n^k \hat{P}_n^{(\alpha+k,\beta+k)}, \quad (3.11)$$

additionally

$$A_{\alpha+k-1,\beta+k-1} B_{\alpha+k-1,\beta+k-1} \hat{P}_n^{(\alpha+k,\beta+k)} = E_{n+1}^{k-1} \hat{P}_n^{(\alpha+k,\beta+k)}, \quad (3.12)$$

in which

$$E_n^{k-1} = (n-1)(n+\alpha+\beta+2k-2). \quad (3.13)$$

The above equations suggest that

$$B_{\alpha+k,\beta+k} A_{\alpha+k,\beta+k} - E_n^k = A_{\alpha+k-1,\beta+k-1} B_{\alpha+k-1,\beta+k-1} - E_{n+1}^{k-1}, \quad (3.14)$$

where ($k = 1, 2, 3, \dots$). Therefore, from Eq. (3.14) it becomes evident that the SUSY pair of Hamiltonians $B_{\alpha+k-1,\beta+k-1} A_{\alpha+k-1,\beta+k-1}$ and $A_{\alpha+k-1,\beta+k-1} B_{\alpha+k-1,\beta+k-1}$ are shape invariant

with a shift only.

We can obtain the X_1 -Jacobi polynomials from the above generalize operators as

$$2^{(k-1)}(k-1)!\hat{P}_k^{(\alpha,\beta)} = B_{\alpha,\beta}B_{\alpha+1,\beta+1}\dots B_{\alpha+k-2,\beta+k-2}\hat{P}_1^{(\alpha+k-1,\beta+k-1)}. \quad (3.15)$$

Let us define a sequence of Hamiltonians as

$$H^k = B_{\alpha+k-1,\beta+k-1}A_{\alpha+k-1,\beta+k-1} + (k-1)(k+\alpha+\beta).$$

It is evident that the SI property is satisfied by this sequence of Hamiltonians, allowing them to be written as,

$$H^{k+1} = H^k + (\alpha + \beta + 2k) = H^k + R_k, \quad (3.16)$$

where $R_k = (\alpha + \beta + 2k)$.

Consistent with our expectations, the n^{th} eigenvalue of the original Hamiltonian H^1 is found to be

$$\begin{aligned} E_n &= \sum_{k=1}^{n-1} (\alpha + \beta) + 2k \\ &= (n-1)(n + \alpha + \beta). \end{aligned} \quad (3.17)$$

3.1.2 X_1 -Laguerre polynomials

For any non-negative integer n ($n \geq 1$) and a positive real number k , the differential equation for the Laguerre polynomials $\hat{L}_n^{(k)}(x)$ is given in Eq. (2.35). This differential equation can be expressed in the following eigenvalue form,

$$\left(-x \frac{d^2}{dx^2} + \frac{(x-k)}{(x+k)} \left[(x+k+1) \frac{d}{dx} - 1 \right] - (n-1) \right) \hat{L}_n^{(k)} = 0, \quad (3.18)$$

This can be written simply as

$$[H - (n-1)] \hat{L}_n^{(k)} = 0, \quad (3.19)$$

where

$$H = -x \frac{d^2}{dx^2} + \frac{(x-k)}{(x+k)} \left[(x+k+1) \frac{d}{dx} - 1 \right]. \quad (3.20)$$

Lowering (A_k) and raising (B_k) operators defined for X_1 -Laguerre polynomials are

$$A_k = -\frac{(x+k+1)^2}{x+k} \frac{d}{dx} \left(\frac{1}{x+k+1} \right), \quad (3.21a)$$

$$B_k = \frac{x(x+k)}{(x+k+1)} \left(\frac{d}{dx} - 1 \right) + k \quad (3.21b)$$

and these operators hold the following relations

$$A_k \hat{L}_n^{(k)} = \hat{L}_{n-1}^{(k+1)}, \quad (3.22a)$$

$$B_k \hat{L}_n^{(k+1)} = n \hat{L}_{n+1}^{(k)}. \quad (3.22b)$$

Therefore, the Hamiltonian in Eq. (3.20) is expressible as a product of A_k and B_k

$$H = B_k A_k = A_{k-1} B_{k-1} - 1. \quad (3.23)$$

From Eqs. (3.22a) and (3.22b), we can obtain the following

$$B_k A_k \hat{L}_n^{(k)} = E_n^0 \hat{L}_n^{(k)}, \quad (3.24)$$

$$A_k B_k (A_k \hat{L}_n^{(k)}) = E_n^0 (A_k \hat{L}_n^{(k)}) \text{ or } A_k B_k (\hat{L}_{n-1}^{(k+1)}) = E_n^0 (\hat{L}_{n-1}^{(k+1)}),$$

where,

$$E_n = (n-1).$$

This implies that the Hamiltonians $A_k B_k$ and $B_k A_k$ form a SUSY partner pair. To establish the SI property, we construct the sequence of Hamiltonians, $A_{k+r-1} B_{k+r-1}$ and $B_{k+r-1} A_{k+r-1}$, where $r = 1, 2, 3, \dots$. It is straightforward to verify that

$$\begin{aligned} B_{k+r-1} A_{k+r-1} \hat{L}_n^{(k+r-1)} &= E_n^{r-1} \hat{L}_n^{(k+r-1)}, \\ A_{k+r-1} B_{k+r-1} (A_{k+r-1} \hat{L}_n^{(k+r-1)}) &= E_n^{r-1} (A_{k+r-1} \hat{L}_n^{(k+r-1)}), \\ \text{or } A_{k+r-1} B_{k+r-1} \hat{L}_{n-1}^{(k+r)} &= E_n^{r-1} \hat{L}_{n-1}^{(k+r)}, \end{aligned} \quad (3.25)$$

with $E_n^{r-1} = (n-1)$.

By using the recurrence relations, we can express

$$\begin{aligned} &B_{k+r} A_{k+r} - (n-1) \\ &= A_{k+r-1} B_{k+r-1} - n. \end{aligned} \quad (3.26)$$

The above equation shows that SUSY partner Hamiltonians in Eq. (3.25) possess SI with a constant shift independent of the parameter r .

A relationship between X_1 -Laguerre polynomials can be established using raising operators, as expressed here

$$(r-1)!\hat{L}_r^{(k)} = B_k B_{k+1} B_{k+2} \dots B_{k+r-2} \hat{L}_1^{(k+r-1)}. \quad (3.27)$$

Now, we introduce the following sequence of Hamiltonians

$$H^r = B_{k+r-1} A_{k+r-1} + (r-1). \quad (3.28)$$

The sequence of Hamiltonians exhibiting SI, also satisfies

$$\begin{aligned} & A_{k+r-1} B_{k+r-1} + (r-1) \\ &= B_{k+r} A_{k+r} + 1 + r - 1 \\ &= B_{k+r} A_{k+r} + r \\ &= H^{r+1} = H^r + 1. \end{aligned} \quad (3.29)$$

Therefore, the n^{th} eigenvalue of the original Hamiltonian in Eq. (3.20) is given by

$$E_n = \sum_{k=1}^{n-1} 1 = (n-1). \quad (3.30)$$

3.2 X_m -Exceptional orthogonal polynomials

In previous section we have seen, how the SUSY and SI are responsible for the solubility of X_1 -Jacobi and X_1 -Laguerre EOPs. In this section we generalize our study for X_m EOPs for any $m \geq 1$.

3.2.1 X_m -Jacobi orthogonal polynomials

The differential equation governing the X_m -Jacobi orthogonal polynomials $\hat{P}_{n,m}^{(\alpha,\beta)}(x)$ is given in Eq. (2.40), where $P_m^{(\alpha,\beta)}(x)$ denotes the classical Jacobi polynomials.

The X_m -Jacobi differential equation can be represented by following eigenvalue equation

$$\begin{aligned} & \left[(x^2 - 1) \frac{d^2}{dx^2} + (x^2 - 1) \left((\alpha - \beta - m + 1) \frac{P_{m-1}^{(-\alpha, \beta)}(x)}{P_m^{(-\alpha-1, \beta-1)}(x)} - \left(\frac{\alpha+1}{1-x} \right) + \left(\frac{\beta+1}{1+x} \right) \right) * \frac{d}{dx} \right. \\ & + \left(\beta(\alpha - \beta - m + 1)(x - 1) \frac{P_{m-1}^{(-\alpha, \beta)}(x)}{P_m^{(-\alpha-1, \beta-1)}(x)} \right. \\ & \left. \left. - m(\alpha - \beta - m + 1) - (n - m)(\alpha + \beta + n - m + 1) \right) \right] \hat{P}_{n,m}^{(\alpha, \beta)}(x) = 0. \end{aligned} \quad (3.31)$$

We can express this formally as

$$[H - (n - m)(\alpha + \beta + n - m + 1)] \hat{P}_{n,m}^{(\alpha, \beta)}(x) = 0, \quad (3.32)$$

where

$$\begin{aligned} H = & (x^2 - 1) \frac{d^2}{dx^2} + (x^2 - 1) \left((\alpha - \beta - m + 1) \frac{P_{m-1}^{(-\alpha, \beta)}(x)}{P_m^{(-\alpha-1, \beta-1)}(x)} - \left(\frac{\alpha+1}{1-x} \right) + \left(\frac{\beta+1}{1+x} \right) \right) \\ & \times \frac{d}{dx} + \left(\beta(\alpha - \beta - m + 1)(x - 1) \frac{P_{m-1}^{(-\alpha, \beta)}(x)}{P_m^{(-\alpha-1, \beta-1)}(x)} - m(\alpha - \beta - m + 1) \right). \end{aligned} \quad (3.33)$$

The lowering and raising operators for the X_m -Jacobi polynomials are

$$A_{\alpha, \beta}^m = \frac{P_m^{(-\alpha-2, \beta)}(x)}{P_m^{(-\alpha-1, \beta-1)}(x)} \left[\frac{d}{dx} - \frac{1}{2}(\beta - \alpha + m - 1) \frac{P_{m-1}^{(-\alpha-1, \beta+1)}(x)}{P_m^{(-\alpha-2, \beta)}(x)} \right], \quad (3.34)$$

and

$$\begin{aligned} B_{\alpha, \beta}^m = & (1 - x^2) \frac{P_m^{(-\alpha-1, \beta-1)}(x)}{P_m^{(-\alpha-2, \beta)}(x)} \left[\frac{d}{dx} - \left(\frac{1}{2}(\beta - \alpha + m - 1) \frac{P_{m-1}^{(-\alpha, \beta)}(x)}{P_m^{(-\alpha-1, \beta-1)}(x)} \right. \right. \\ & \left. \left. + \left(\frac{\alpha+1}{1-x} \right) - \left(\frac{\beta+1}{1+x} \right) \right) \right]. \end{aligned} \quad (3.35)$$

These lowering and raising operators hold the following relations

$$A_{\alpha, \beta}^m \hat{P}_{n,m}^{(\alpha, \beta)}(x) = \frac{1}{2}(\alpha + \beta + n - m + 1) \hat{P}_{n-1,m}^{(\alpha+1, \beta+1)}(x) \quad n \geq m \quad (3.36)$$

and

$$B_{\alpha, \beta}^m \hat{P}_{n,m}^{(\alpha+1, \beta+1)}(x) = 2(n - m + 1) \hat{P}_{n+1,m}^{(\alpha, \beta)}(x) \quad n \geq m. \quad (3.37)$$

It is now possible to factorize the Hamiltonian in (3.33) using the mentioned relations as follows

$$H = B_{\alpha,\beta}^m A_{\alpha,\beta}^m = A_{\alpha-1,\beta-1}^m B_{\alpha-1,\beta-1}^m - (\alpha + \beta). \quad (3.38)$$

Based on Eqs. (3.36) and (3.37), we obtain

$$\begin{aligned} B_{\alpha,\beta}^m A_{\alpha,\beta}^m \hat{P}_{n,m}^{(\alpha,\beta)}(x) &= E_{n,m}^0 \hat{P}_{n,m}^{(\alpha,\beta)}(x), \\ A_{\alpha,\beta}^m B_{\alpha,\beta}^m (A_{\alpha,\beta}^m \hat{P}_{n,m}^{(\alpha,\beta)}(x)) &= E_{n,m}^0 (A_{\alpha,\beta}^m \hat{P}_{n,m}^{(\alpha,\beta)}(x)), \end{aligned} \quad (3.39)$$

where

$$E_{n,m}^0 = (n - m)(\alpha + \beta + n - m + 1). \quad (3.40)$$

Therefore, the Hamiltonians

$$\begin{aligned} H^1 &= B_{\alpha,\beta}^m A_{\alpha,\beta}^m, \\ H^2 &= A_{\alpha,\beta}^m B_{\alpha,\beta}^m, \end{aligned} \quad (3.41)$$

are confirmed to be SUSY partners. We extend our analysis by introducing a parameter k . This allows us to construct a sequence of Hamiltonians. Notably, for all positive integer values of $k(= 1, 2, 3, \dots)$, the pairs of Hamiltonians $B_{\alpha+k-1,\beta+k-1}^m A_{\alpha+k-1,\beta+k-1}^m$ and $A_{\alpha+k-1,\beta+k-1}^m B_{\alpha+k-1,\beta+k-1}^m$ also exhibit SUSY partner behavior.

$$\begin{aligned} B_{\alpha+k-1,\beta+k-1}^m A_{\alpha+k-1,\beta+k-1}^m |\psi^k\rangle &= E_{n,m}^{k-1} |\psi^k\rangle \\ A_{\alpha+k-1,\beta+k-1}^m B_{\alpha+k-1,\beta+k-1}^m (A_{\alpha+k-1,\beta+k-1}^m |\psi^k\rangle) &= E_{n,m}^{k-1} (A_{\alpha+k-1,\beta+k-1}^m |\psi^k\rangle) \end{aligned} \quad (3.42)$$

where, $k = 1, 2, 3, \dots$ and

$$E_{n,m}^{k-1} = (n - m)(\alpha + \beta + n - m + 2k - 1). \quad (3.43)$$

it can be written as

$$B_{\alpha+k,\beta+k}^m A_{\alpha+k,\beta+k}^m \hat{P}_{n,m}^{(\alpha+k,\beta+k)}(x) = E_{n,m}^k \hat{P}_{n,m}^{(\alpha+k,\beta+k)}(x), \quad (3.44)$$

and

$$A_{\alpha+k-1,\beta+k-1}^m B_{\alpha+k-1,\beta+k-1}^m \hat{P}_{n,m}^{(\alpha+k,\beta+k)}(x) = E_{n+1,m}^{k-1} \hat{P}_{n,m}^{(\alpha+k,\beta+k)}(x). \quad (3.45)$$

By using Eqs. (3.44) and (3.45), we can express,

$$B_{\alpha+k,\beta+k}^m A_{\alpha+k,\beta+k}^m = A_{\alpha+k-1,\beta+k-1}^m B_{\alpha+k-1,\beta+k-1}^m + E_{n,m}^k - E_{n+1,m}^{k-1}, \quad (3.46)$$

here ($k=1,2,3,\dots$). From Eq. (3.46), we can conclude that the SUSY partner Hamiltonians

defined in Eq. (3.42) exhibit a remarkable property i.e. they are shape-invariant with only a shift.

From the previously mentioned generalized operators, we can express the X_m Jacobi polynomials as

$$2^{(k-m)}(k-m)!\hat{P}_{k,m}^{(\alpha,\beta)}(x) = B_{\alpha,\beta}^m B_{\alpha+1,\beta+1}^m \dots B_{\alpha+k-m-1,\beta+k-m-1}^m \hat{P}_{m,m}^{(\alpha+k-m,\beta+k-m)}(x). \quad (3.47)$$

We now define a family of Hamiltonians

$$H^k = B_{\alpha+k-1,\beta+k-1}^m A_{\alpha+k-1,\beta+k-1}^m + (k-1)(k+\alpha+\beta). \quad (3.48)$$

We can see that these shape invariant Hamiltonians satisfy

$$\begin{aligned} & A_{\alpha+k-1,\beta+k-1}^m B_{\alpha+k-1,\beta+k-1}^m + (k-1)(k+\alpha+\beta) \\ &= B_{\alpha+k,\beta+k}^m A_{\alpha+k,\beta+k}^m + (\alpha+\beta+2k) + (k-1)(k+\alpha+\beta). \end{aligned} \quad (3.49)$$

Therefore, we can express

$$H^{k+1} = H^k + R_k, \quad (3.50)$$

here $R_k = (\alpha + \beta + 2k)$.

Therefore, the n^{th} eigenvalue of Hamiltonian H^1 is

$$\begin{aligned} E_n^m &= \sum_{k=1}^{n-m} (\alpha + \beta) + 2k \\ &= (n-m)(\alpha + \beta + n - m + 1). \end{aligned} \quad (3.51)$$

3.2.2 X_m -Laguerre orthogonal polynomials

For given non-negative integer $m \geq 0$ and $n \geq 1$, the X_m -Laguerre orthogonal polynomials $\hat{L}_{n,m}^k(x)$ satisfy the differential equation given in Eq. (2.48), where $L_m^k(x)$ denote the classical Laguerre polynomials.

We can express X_m -Laguerre differential equation in the form of eigenvalue equation as

$$\begin{aligned} & \left[-x \frac{d^2}{dx^2} + \left((x-k-1) + 2x \frac{L_{m-1}^k(-x)}{L_m^{k-1}(-x)} \right) \frac{d}{dx} \right. \\ & \quad \left. + \left(2k \frac{L_{m-1}^k(-x)}{L_m^{k-1}(-x)} - n \right) \right] \hat{L}_{n,m}^k(x) = 0. \end{aligned} \quad (3.52)$$

We can express the above equation in a formal way as,

$$[H - (n - m)]\hat{L}_{n,m}^k(x) = 0, \quad (3.53)$$

where

$$H = -x \frac{d^2}{dx^2} + \left((x - k - 1) + 2x \frac{L_{m-1}^k(-x)}{L_m^{k-1}(-x)} \right) \frac{d}{dx} + \left(2k \frac{L_{m-1}^k(-x)}{L_m^{k-1}(-x)} - m \right). \quad (3.54)$$

The lowering and raising operators for the X_m -Laguerre orthogonal polynomials that allow the degree of the polynomials to be decreased or increased while preserving orthogonality are given by specific relations

$$A_k^m = -\frac{L_m^k(-x)}{L_m^{k-1}(-x)} \left(\frac{d}{dx} - \frac{L_{m-1}^{k+1}(-x)}{L_m^k(-x)} \right), \quad (3.55)$$

and

$$B_k^m = \frac{L_m^{k-1}(-x)}{L_m^k(-x)} \left(x \frac{d}{dx} + (1 + k) \right) - x. \quad (3.56)$$

These above operators satisfy the following relations

$$A_k^m \hat{L}_{n,m}^k(x) = \hat{L}_{n-1,m}^{k+1}(x), \quad (3.57)$$

and

$$B_k^m \hat{L}_{n,m}^{k+1}(x) = (n - m + 1) \hat{L}_{n+1,m}^k(x), \quad (3.58)$$

Utilizing these raising and lowering operators, we can express the Hamiltonian in Eq. (3.54) in a factorized form as

$$H = B_k^m A_k^m = A_{k-1}^m B_{k-1}^m - 1. \quad (3.59)$$

Following the Eqs. (3.57) and (3.58), we can write

$$\begin{aligned} B_k^m A_k^m \hat{L}_{n,m}^k(x) &= E_{n,m} \hat{L}_{n,m}^k(x), \\ A_k^m B_k^m (A_k^m \hat{L}_{n,m}^k(x)) &= E_{n,m} (A_k^m \hat{L}_{n,m}^k(x)), \end{aligned} \quad (3.60)$$

where $E_{n,m} = (n - m)$.

From the above equation (3.60), it can be concluded that Hamiltonians $A_k^m B_k^m$ and $B_k^m A_k^m$ exhibit SUSY partner behavior.

Now, we extend our analysis by introducing a parameter, denoted by r and for all positive integer values of r (i.e., $r = 1, 2, 3, \dots$), the pairs of Hamiltonians $A_{k+r-1}^m B_{k+r-1}^m$ and

$B_{k+r-1}^m A_{k+r-1}^m$ exhibit SUSY partner behavior, provided by the conditions,

$$\begin{aligned} B_{k+r-1}^m A_{k+r-1}^m \hat{L}_{n,m}^{k+r-1}(x) &= E_{n,m}^{r-1} \hat{L}_{n,m}^{k+r-1}(x) \\ A_{k+r-1}^m B_{k+r-1}^m (A_{k+r-1}^m \hat{L}_{n,m}^{k+r-1}(x)) &= E_{n,m}^{r-1} (A_{k+r-1}^m \hat{L}_{n,m}^{k+r-1}(x)), \end{aligned} \quad (3.61)$$

where $E_{n,m}^{r-1} = (n - m)$.

We observe that the eigenvalues of the Hamiltonians exhibit no dependence on the parameter r . As a result, all n eigenstates share the same degenerate eigenvalues.

The recurrence relation yields the following form

$$\begin{aligned} &B_{k+r}^m A_{k+r}^m - (n - m) \\ &= A_{k+r-1}^m B_{k+r-1}^m - (n - m + 1). \end{aligned} \quad (3.62)$$

From the above relation, we can conclude that the SUSY partner Hamiltonians introduced in Eq. (3.61) exhibit SI, characterized by a shift.

The X_m -Laguerre polynomials can also be related through raising operators as demonstrated below

$$(r - m)! \hat{L}_{r,m}^{(k)} = B_k^m B_{k+1}^m B_{k+2}^m \cdots B_{k+r-m-1}^m \hat{L}_1^{(k+r-m)}. \quad (3.63)$$

Now, we define the family of Hamiltonians

$$H^r = B_{k+r-1}^m A_{k+r-1}^m + (r - 1). \quad (3.64)$$

We find that this family of shape-invariant Hamiltonians satisfy the condition

$$\begin{aligned} &A_{k+r-1}^m B_{k+r-1}^m + (r - 1) \\ &= B_{k+r}^m A_{k+r}^m + r, \end{aligned} \quad (3.65)$$

or, this relationship can be expressed as $H^{r+1} = H^r + 1$, which directly leads to the n^{th} eigenvalue of the Hamiltonian in Eq. (3.54)

$$E_n^m = \sum_{k=1}^{n-m} 1 = (n - m). \quad (3.66)$$

3.3 Conclusion

Exceptional Laguerre and Jacobi polynomials play a crucial role in the rational extension of various quantum mechanical problems, making them worthy of detailed investigation. In this paper, we present a novel approach to examining the differential equations governing these EOPs and their solutions within the framework of SUSY and SI. We explicitly

demonstrate the SUSY and SI properties of the Hamiltonians associated with X_1 -Jacobi and X_1 -Laguerre polynomials. The underlying SI symmetry is shown to be the key factor responsible for the solubility of the differential equations corresponding to these EOPs. We further generalize these results to X_m -Jacobi and X_m -Laguerre polynomials for arbitrary values of m .